

1. $\int_0^1 \int_0^x xy \, dy \, dx =$

- (A) 0 (B) $1/8$ (C) $1/3$ (D) 1 (E) 3

2. If D is the disk $x^2 + y^2 \leq 4$, then $\iint_D e^{-(x^2+y^2)} \, dA =$

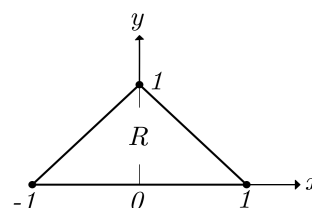
- (A) 4π (B) πe^{-4} (C) $4\pi e^{-4}$ (D) $\pi(1 - e^{-4})$ (E) $4\pi(e - e^{-4})$

3. The point $(x, y, z) = (-1, 1, 1)$ is written in cylindrical coordinates as

- (A) $(\sqrt{2}, 3\pi/4, 1)$ (B) $(\sqrt{2}, 7\pi/4, 1)$ (C) $(\sqrt{3}, 3\pi/4, 1)$
 (D) $(\sqrt{3}, \pi/4, \pi/4)$ (E) $(-1, \sqrt{2}, \pi/4)$

4. The point on the plane $2x + y + 3z = 3$ that is closest to the origin is

- (A) $(0, 0, 1)$ (B) $(3/7, 3/14, 9/14)$ (C) $(7/15, 8/15, 1/15)$
 (D) $(5/6, 1/3, 1/3)$ (E) $(1, 1, 1/3)$



5. If R is the triangular region shown above, then $\iint_R f(x, y) \, dA =$

- (A) $\int_{-1}^1 \int_0^1 f(x, y) \, dy \, dx$ (B) $\int_0^\pi \int_0^1 f(r \cos \theta, r \sin \theta) r \, dr \, d\theta$ (C) $\int_{-1}^0 \int_0^{x+1} f(x, y) \, dx \, dy$
 (D) $\int_0^1 \int_0^{1-x} f(x, y) \, dy \, dx$ (E) $\int_0^1 \int_{y-1}^{1-y} f(x, y) \, dx \, dy$

6. (15 pts) Let S be the surface in the first octant given by $xyz = 1$. Find a point on S which minimizes $f(x, y, z) = x + 2y + 3z$.

7. (15 pts) Let E be the solid that lies inside the cone $z = \sqrt{3x^2 + 3y^2}$, outside the sphere $x^2 + y^2 + z^2 = 1$, and below the plane $z = 10$. If the density of E is $f(x, y, z) = x^2 + y^2$, express the mass of E as a triple iterated integral in spherical coordinates (do not evaluate).

8. (20 pts) Evaluate the integral by reversing the order of integration

$$\int_0^1 \int_{\arcsin y}^{\pi/2} \cos x \sqrt{1 + \cos^2 x} \, dx \, dy$$

9. (20 pts) Let D be the region inside the circle $x^2 + y^2 = 2x$ and outside the circle $x^2 + y^2 = 1$.

Evaluate $\iint_D (x^2 + y^2)^{-1/2} \, dA$.

$$1. \int_0^1 \int_0^x xy \, dy \, dx = \int_0^1 x \left. \frac{y^2}{2} \right|_{y=0}^{y=x} dx = \int_0^1 \frac{1}{2} x^3 dx = \frac{1}{2} \left. \frac{x^4}{4} \right|_0^1 = \frac{1}{8} \quad (B)$$

$$2. \int_0^{2\pi} \int_0^2 e^{-r^2} r \, dr \, d\theta = \int_0^{2\pi} d\theta \cdot \left(-\frac{1}{2} \right) e^{-r^2} \Big|_{r=0}^{r=2} = \pi(1 - e^{-4}) \quad (D)$$

$$3. \left. \begin{array}{l} x = -1 \\ y = 1 \end{array} \Rightarrow \begin{array}{l} r = \sqrt{2} \text{ \& } \tan \theta = -1 \\ \text{\& } (x, y) \text{ in second quadrant} \end{array} \right\} \Rightarrow \theta = \frac{3\pi}{4} \Rightarrow \text{cylindrical coordinates } (r, \theta, z) = (\sqrt{2}, \frac{3\pi}{4}, 1) \quad (A)$$

$$4. f(x, y, z) = x^2 + y^2 + z^2, \quad g(x, y, z) = 2x + y + 3z - 3$$

$$\left\{ \begin{array}{l} \nabla f = \lambda \nabla g \\ g = 0 \end{array} \right\} \Rightarrow \left\{ \begin{array}{l} 2x = 2\lambda \\ 2y = \lambda \\ 2z = 3\lambda \\ 2x + y + 3z = 3 \end{array} \right. \Rightarrow \left\{ \begin{array}{l} x = \lambda \\ y = \lambda/2 \\ z = 3\lambda/2 \\ (2 + \frac{1}{2} + \frac{9}{2})\lambda = 3 \end{array} \right. \Rightarrow \lambda = 3/7 \text{ \& } (x, y, z) = (\frac{3}{7}, \frac{3}{14}, \frac{9}{14}) \quad (B)$$

$$5. \iint_R f \, dA = \int_0^1 \int_{y-1}^{1-y} f(x, y) \, dx \, dy \quad (B)$$

$$6. f(x, y, z) = x + 2y + 3z, \quad g(x, y, z) = xyz - 1$$

$$\left\{ \begin{array}{l} \nabla f = \lambda \nabla g \\ g = 0 \end{array} \right\} \Rightarrow \left\{ \begin{array}{l} 1 = \lambda yz \\ 2 = \lambda xz \\ 3 = \lambda xy \\ xyz = 1 \end{array} \right. \Rightarrow \left\{ \begin{array}{l} \frac{1}{2} = \frac{y}{x} \\ \frac{1}{3} = \frac{z}{x} \\ xyz = 1 \end{array} \right. \Rightarrow x \cdot \frac{x}{2} \cdot \frac{x}{3} = 1$$

$$x^3 = 6 \Rightarrow x = \sqrt[3]{6}$$

$$\Rightarrow \text{solution } (x, y, z) = (\sqrt[3]{6}, \frac{1}{2}\sqrt[3]{6}, \frac{1}{3}\sqrt[3]{6})$$

$$7. \text{Spherical coordinates } \left\{ \begin{array}{l} x = \rho \sin \phi \cos \theta \\ y = \rho \sin \phi \sin \theta \\ z = \rho \cos \phi \end{array} \right.$$

$$\text{Density: } f = x^2 + y^2 = r^2 = \rho^2 \sin^2 \phi$$

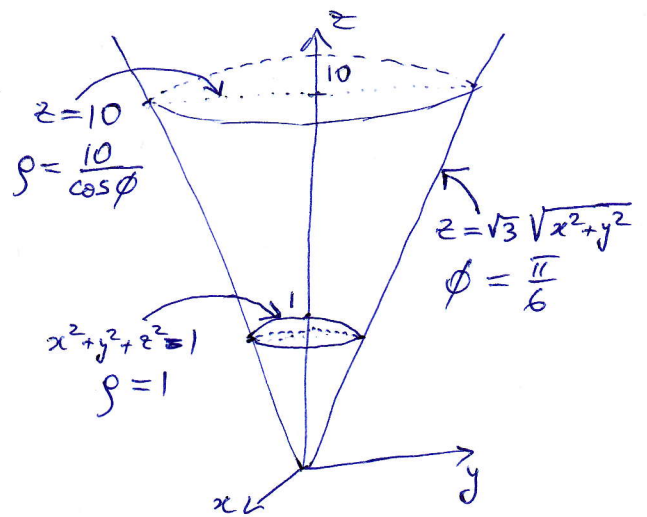
$$\text{Cone: } z = \sqrt{3(x^2 + y^2)} = \sqrt{3} r$$

$$\rho \cos \phi = \sqrt{3} \rho \sin \phi \Rightarrow \tan \phi = \frac{1}{\sqrt{3}} \Rightarrow \phi = \pi/6$$

$$\text{Sphere: } x^2 + y^2 + z^2 = 1 \Rightarrow \rho = 1$$

$$\text{Plane: } z = 10 \Rightarrow \rho \cos \phi = 10$$

$$\rho = 10 \sec \phi$$



$$\text{Mass} = \iiint_E f \, dV = \int_0^{2\pi} \int_0^{\pi/6} \int_1^{10 \sec \phi} \rho^2 \sin^2 \phi \, \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

$$= \int_0^{2\pi} \int_0^{\pi/6} \int_1^{10 \sec \phi} \rho^4 \sin^3 \phi \, d\rho \, d\phi \, d\theta$$

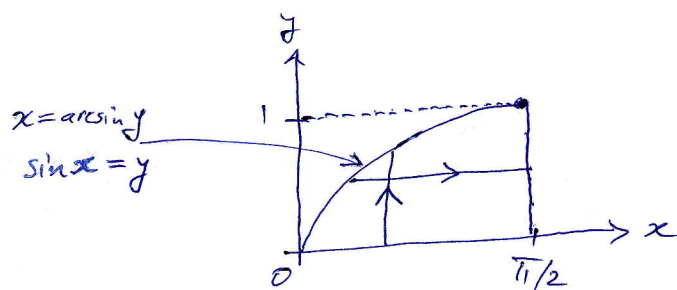
$$8. \int_0^1 \int_{\arcsin y}^{\pi/2} \cos x \sqrt{1+\cos^2 x} \, dx \, dy$$

$$= \int_0^{\pi/2} \int_0^{\sin x} \cos x \sqrt{1+\cos^2 x} \, dy \, dx$$

$$= \int_0^{\pi/2} \sin x \cos x \sqrt{1+\cos^2 x} \, dx$$

$$= -\frac{1}{2} \frac{(1+\cos^2 x)^{3/2}}{3/2} \Big|_0^{\pi/2}$$

$$= -\frac{1}{3} \left(1^{\frac{3}{2}} - 2^{\frac{3}{2}} \right) = \frac{\sqrt{8}-1}{3}$$



$$u = 1 + \cos^2 x$$

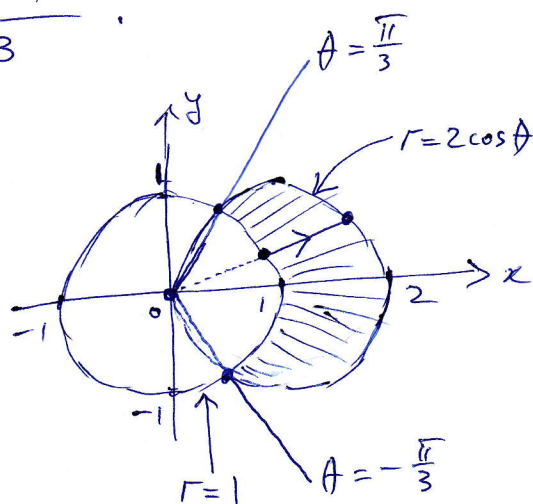
$$du = -2 \cos x \sin x \, dx$$

9. Polar coordinates: $x = r \cos \theta$
 $y = r \sin \theta$

Circle $x^2 + y^2 = 1 \Rightarrow r = 1$

Circle $x^2 + y^2 = 2x \Rightarrow r^2 = 2r \cos \theta$
or $(x-1)^2 + y^2 = 1 \quad r = 2 \cos \theta$

Intersection: $x^2 + y^2 = 1 = 2x$
 $\Rightarrow r = 1 = 2 \cos \theta \Rightarrow \theta = \pm \frac{\pi}{3}$



$$\iint_D (x^2 + y^2)^{-1/2} \, dA = \int_{-\pi/3}^{\pi/3} \int_1^{2 \cos \theta} (r^2)^{-1/2} r \, dr \, d\theta$$

$$= \int_{-\pi/3}^{\pi/3} \int_1^{2 \cos \theta} dr \, d\theta = 2 \int_0^{\pi/3} (2 \cos \theta - 1) \, d\theta$$

$$= 2 \left[2 \sin \theta - \theta \right]_0^{\pi/3} = 2 \left(\sqrt{3} - \frac{\pi}{3} \right)$$